

Success Criteria Matrix

ROV for High Temperature (ROCK)

ECE 129A

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1. Project Criteria Overview

The ROV for High Temperature (ROCK) project aims to design and prototype a remotely operated vehicle that can safely enter hazardous, high-temperature environments while an operator remains at a computer-based user interface connected through a CAT6a tether. The ROV will carry front and rear cameras, a cluster of gas sensors for CO₂, CO, H₂S, and CH₄, as well as temperature, barometric pressure, and humidity sensors that all feed into an STM32 microcontroller and a Raspberry Pi for data processing and video streaming. Success at a high level means that the system provides reliable situational awareness and environmental measurements, maintains stable power and communication through the AC supply, rectifier, and buck converter chain, and allows an operator to maneuver the ROV and interpret sensor readings without confusion or unsafe behavior. A successful outcome is a prototype that performs these functions consistently in representative test environments while meeting the structural, electrical, usability, and cost constraints agreed upon by the team and stakeholders.

2. Success Criteria Matrix

Criterion	Metric	Target / Goal	How We Will Measure It
Structural and thermal durability	Chassis and enclosure integrity at elevated temperature	Operates 30 minutes at 50°C with no failures (stretch goal of 60°C)	Place ROV in a 50°C chamber and run motors and sensors continuously; inspect afterwards
Power system endurance	Runtime under typical mission load	≥ 45 minutes	Full-power runtime test with cameras, sensors, and motors active; record until shutdown
Communication robustness	Data link stability over CAT6a tether	$< 2\%$ packet loss and < 150 ms round-trip latency at 30 m	Send timestamped packets over 50 meter CAT6a cable and analyze logs
Gas sensing accuracy	Error for CO ₂ , CO, H ₂ S, CH ₄	$< \pm 10\%$ error compared to references	Test in calibration gas chamber and compare with reference instrumentation
Environmental sensor accuracy	Temperature, pressure, and humidity error	Temp $\pm 2^\circ\text{C}$, Pressure ± 2 hPa, Humidity $\pm 5\%$ RH	Compare sensor readings to calibrated reference probes in a stable environment